Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/03/2020
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Project Title:	Cervical Lymphatic Function Quantification and Associated Molecular Changes in Response to Simulated Microgravity		
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Zip Code:	32502-6008	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 HERO NNJ16ZSA001N-TRIRT. Appendix C: Translational Research Institute for Space Health (TRISH) Research Topics
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 4/30/2020 per TRISH report submission July 2020 (Ed., 9/26/20) NOTE: End date changed to 3/31/2020 per E. Urquieta/TRISH (Ed., 5/23/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Crandall, David Ph.D. (Indiana University School of Medicine)		
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Task Description:	An increased presence of white matter hyperintensities (WMH) has been previously described in astronauts; however, the underlying cause of this pathology is not clearly understood. Astronauts exposed to long-duration microgravity also develop varying degrees of visual change and signs of elevated intracranial pressure (ICP), which are thought to potentially be associated with the increased WMH seen in this population. Central nervous systems (CNS) changes associated with extreme environmental exposures, including increased WMH, have also been detected in high altitude aviators and professional scuba divers. There is growing evidence that a proportion of the cerebrospinal fluid drains from the cranium via the deep cervical lymphatic pathway. Recent literature has also revealed the existence of lymphatic vessels in the brain that serve to regulate fluid pressure and pathological clearance for surrounding tissue, and these vessels are connected to the peripheral lymphatics. Impairment of these drainage systems in other pathological models have demonstrated histological changes, including tau protein accumulation and white matter damage, in the brain. Changes in fluid dynamics with environmental exposures, such as microgravity or immersion, could lead to impairment of brain lymphatic drainage, resulting in accumulation of fluid and elevated ICP, stasis of brain metabolic waste, and subsequent tissue damage and lesion development. At present, however, the effect of microgravity on deep cervical lymphatic function has not yet been characterized. Impact: Understanding the effect of microgravity on cervical lymphatic function and outflow from the brain will provide insight into factors and potential mitigation strategies for dangerous training or mission-limiting physiological conditions, such as increased ICP and development of WMH, which could detrimentally impact astronaut performance both acutely and long-term. In addition, further refinement of lymphatic function monitoring technology could lead to its utilization in space as a diagnostic tool for crew members experiencing CNS symptoms. The proposed study will address the following Specific Aims: Specific Aim 1: Refine the current near-infrared fluorescence lymphatic imaging (NIRFLI) system underwater lymphatic imaging. Specific Aim 2: Characterize and compare the effect of deep cervical lymphatic function in humans in response to two models of simulated microgravity: head-out immersion (HOI) and head-down tilt (HDT). Specific Aim 3: Develop a visual quantification software program for objectively measuring lymphatic function as captured by the device. Specific Aim 4: Evaluate targeted molecular biomarkers associated with impaired immune function and lymphatic clearance in response to the HOI and HDT models of simulated microgravity. Approach: The proposed study will refine existing lymphatic imaging technology while addressing the hypothesis that simulated microgravity (HDT, HOI) in humans impairs lymphatic transit speed in the head/ neck lymphatics as compared to baseline due to the associated increase in hydrostatic pressure on the lymphatic system while increasing lymphatic transit in a chest- and head-ward direction. Study participants will undergo dry baseline measurements of cervical lymphatic function using a novel, portable, low footprint method for assessing lymphatic function, followed by measurements conducted during and following simulated microgravity exposures. Each subject will undergo both HDT and HOI, separated by at least 48 hours. Venous blood samples will be collected for assessment of targeted biomarkers prior to baseline measurements and at the conclusion of each exposure. In parallel with study data collection, visual analysis and machine learning algorithms will be applied to develop a software program that automatically quantifies lymphatic function in conjunction with the measuring device.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project characterized the effect of simulated microgravity on the cervical lymphatic system through novel application of existing technology, identifying the thoracic duct as a potential structure to be measured in future ground analog and spaceflght studies assessing headward fluid shift. This project also validated technology that could be used in-flight for research and, eventually, for monitoring crew health. We developed a detailed training and testing protocol for using the Butterfly ultrasound device to measure head and neck structures that are critical for assessing headward fluid shift, including optic nerve sheath diameter and internal jugular vein diameter. This protocol could be used for spaceflight research, and we plan to apply this protocol to other extreme environment research studies in the field and in the laboratory. In addition to future space medicine research and spaceflight application for crew health monitoring, the head and neck protocol could be used clinically for testing patients with cerebrospinal fluid (CSF) or lymphatic disorders. Specifically, the low footprint Butterfly iQ ultrasound is ideal for field applications, and the established protocol could be used for remote assessment of individuals who potentially have impaired CSF outflow. These potential applications would require further validation in the clinical setting and against standard neuroimaging.
	1. Original Project Aims and Objectives Understanding the effect of microgravity on cervical lymphatic function could provide insight into factors and potential mitigation strategies for dangerous training or mission-limiting physiological conditions, which could detrimentally impact astronaut performance both acutely and long-term. Further development of lymphatic function monitoring technology could also be utilized in space as a diagnostic tool for crew members. The proposed study will address the following Specific Aims: Specific Aim 1: Characterize the response of lymphatic function humans in response to the head-out immersion and head-down tilt models of simulated microgravity. Specific Aim 2: Develop a visual quantification software program for objectively measuring lymphatic function based on images captures by the device. Specific Aim 3: Evaluate targeted peripheral biomarkers to begin characterization of impaired brain lymphatic clearance at the molecular level. 2. Key Findings a) Three technologies were assessed; the third technology (Butterfly iQ ultrasound) was utilized for study data collection

Task Progress:	due to field/mission-feasibility. b) A novel approach for measuring cervical lymphatic status was developed and tested. c) A Butterfly ultrasound protocol was developed for head and neck measurements of lymphatics, internal jugular vein (IJV), and optic nerve sheath diameter (OSND) measurement. This protocol includes shared lessons-learned and training material for imaging these structures. d) Inter- and intra-rater reliability assessment applied to Butterfly protocol measurements demonstrated minimal intra- and inter-rater variability with proper training and practice. e) Simulated microgravity (head-down tilt and head-out immersion) study data collection was completed; study results demonstrated significant differences in OSND and IJV when comparing baseline to head-down tilt or head-out immersion. Although there was a visible difference from baseline to experimental condition in thoracic duct volume, total thoracic duct measurements were smaller in number compared to IJV and OSND and will require a larger study number in the future to verify if these changes are statistically significant. f) The study team was only able to consistently capture cervical lymph node volume in two participants across all study conditions. Cervical lymph nodes are highly variable in accessibility for imaging, but future portable ultrasound technology with deeper penetration and improved resolution could be useful for imaging lymph nodes using the protocol developed in this study. g) A Translational Research Institute for Space Health (TRISH)-funded collaborative effort developed an application of machine learning to optimize locating lymphatic structures and optimize image capture. 3. Impact of Key Findings (on hypotheses, technology requirements, objectives, and specific aims of the original proposal--expand across all years) This project characterized the effect of simulated microgravity on the cervical lymphatic system through novel application of existing technology. The initial results from this study suggest impairment of lymphatic outflow by measurement of the thoracic duct at baseline and following exposure to simulated microgravity; however, future studies with a larger number of participants or technology that allows imaging of deeper structures will be important before translating this approach. This project also validated technology that could be used in-flight for research and, eventually, for monitoring crew health. We developed a detailed training and testing protocol for using the Butterfly ultrasound device to measure head and neck structures that are critical for assessing headward fluid shift, including optic nerve sheath diameter and internal jugular vein diameter. In addition to future space medicine research and spaceflight application for crew health monitoring, the head and neck protocol could be used clinically for testing patients with cerebrospinal fluid (CSF) or lymphatic disorders. Specifically, the low footprint Butterfly iQ ultrasound is ideal for field applications, and the established protocol could be used for remote assessment of individuals who potentially have impaired CSF outflow. These potential applications would require further validation in the clinical setting and against standard neuroimaging. 4. Proposed Research Plan for the Future At the time of study completion, our team has been funded for a new study to assess the effect of extreme environmental conditions (e.g., hypercapnia and hypoxia) on brain lymphatic function as imaged by magnetic resonance imaging (MRI). We will also collect secondary measurements to assess lymphatic structures, IJV, and OSND using the protocol/findings from this study, and we will compare changes observed by ultrasound to changes observed by MRI.
Bibliography Type:	Description: (Last Updated: 01/09/2019)